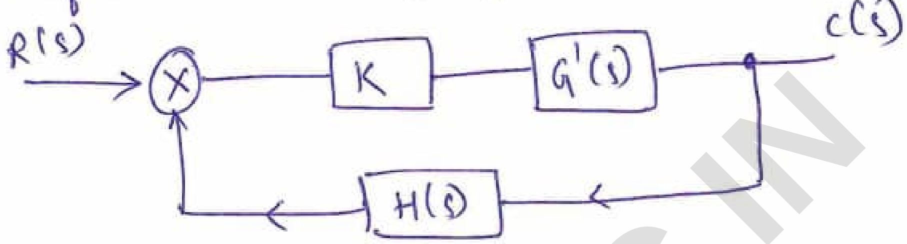
Root Locus

In general, the characteristic equation of a closed loop system is given by,

$$1 + G(s)H(s) = 0$$

For root locus, the gain 'K' is assumed to be a variable parameter and is a part of forward path of the closed loop system.



where $G(s) = K G'(s)$

& K is gain of the system in forward path.

$\therefore$ The characteristic equation becomes,

$$1 + G(s)H(s) = 0$$

i.e $$1 + KG'(s)H(s) = 0$$

If now gain 'K' is varied from $-\infty$ to $+\infty$ then for each separate value of 'K' we will get separate set of locations of the roots of the characteristic equation. If all such locations are joined, the resulting locus is called Root Locus.

One can define root locus as, the locus of the closed loop poles obtained when system gain 'K' is varied from $-\infty$ to ∞ is called Root Locus.

Rules for construction of Root Locus:

Rule 1: The root locus is always symmetrical about the real axis. Get the general information about number of open loop poles (P), zeros (Z) and numbers of branches (N)

Rule 2: Draw the pole-zero plot.
If $P > Z$, then number of branches equal to P, i.e. $N = P$
Elseif $Z > P$, then number of branches equal to Z, i.e. $N = Z$.

Rule 3: Identify sections of real axis for the existence of the root locus.
A point on the real axis lies on the root locus if the sum of the number of open loop poles and open loop zeros, on the real axis, to the right side of this point is odd.

Rule 4: Calculate angles of asymptotes.
Generally number of poles are more than number of zeros and in such case 'P-Z' branches will approach to infinity. The branches which are approaching to infinity, do so along the straight lines called Asymptotes of the root locus. Angles of such asymptotes are given by,

$$\theta = \frac{(2q+1)180^\circ}{P-Z} \quad \text{where } q = 0, 1, 2 \ldots (P-Z-1)$$

Rule 5: Determine the centroid.
Now, only the angles of asymptotes are not sufficient but where the asymptotes are located in s-plane is equally important. All the asymptotes intersect the real axis at a common point known as Centroid and is denoted by σ and is calculated by,

$$\sigma = \frac{\sum \text{Real parts of poles of } G(s)H(s) - \sum \text{Real parts of zeros of } G(s)H(s)}{P-Z}$$

Rule 6 : Breakaway Point

Breakaway point is a point on the real root locus where multiple roots of the characteristic equation occurs, for a particular value of K.

- If there are adjacently placed poles on the real axis and the real axis between them is a part of the root locus then there exists minimum one breakaway point in between adjacently placed poles.

Steps to determine the breakaway point :

Step1 : Construct the characteristic equation $1+G(s)H(s)=0$ of the system.

Step2 : From this equation, separate the terms involving 'K' and terms involving 's'. Write the value of 'K' in terms of 's'.

$K = F(s)$.

Step3 : Differentiate the equation w.r.t 's', equate it to zero. $\frac{dK}{ds} = 0$

Step4 : Roots the equation $\frac{dK}{ds} = 0$ gives us the breakaway points.

Rule 7 : Intersection of Root Locus with imaginary axis. This can be determined by following procedure.

Step1 : Consider the characteristic equation $1+G(s)H(s)=0$

Step2 : Construct Routh's array in terms of 'K'.

Step3 : Determine Kmarginal i.e value of 'K' which creates one of the rows of Routh's array as row of zeros, except the row of s^0.

Step 4 : Construct auxiliary equation $A(s) = 0$ by using the coefficients of a row which is just above the row of zeros.

Step 5 : Roots of auxiliary equation $A(s) = 0$ for $K = K_{mar}$ are nothing but the intersection points of the root locus with imaginary axis.

Rule 8 : Angle of departure at complex conjugate poles and angle of arrival at complex conjugate zeros.

As branch always leaves from an open loop pole, it is advantageous to know at what angle it departs from complex conjugate pole. This angle at which it departs from complex pole is called as angle of departure denoted as ϕ_d.

$$\phi_d = 180° - \phi \quad \text{where } \phi = \Sigma\phi_p - \Sigma\phi_z$$

where $\Sigma\phi_p$ = Contributions by the angles made by remaining open loop poles at the pole at which ϕ_d is to be calculated.

$\Sigma\phi_z$ = Contributions by the angles made by the open loop zeros at the pole at which ϕ_d is to be calculated.

Similarly, angle of arrival at a complex zero is given by, $\phi_a = 180° + \phi$ where $\phi = \Sigma\phi_p - \Sigma\phi_z$

Rule 9 : Construct the root locus by making use of Rule 1 to Rule 8 and predict the stability of the system.

Numericals:

1. Sketch the complete root locus of the system having

$$G(s)H(s) = \frac{K}{s(s+3)(s+5)}$$

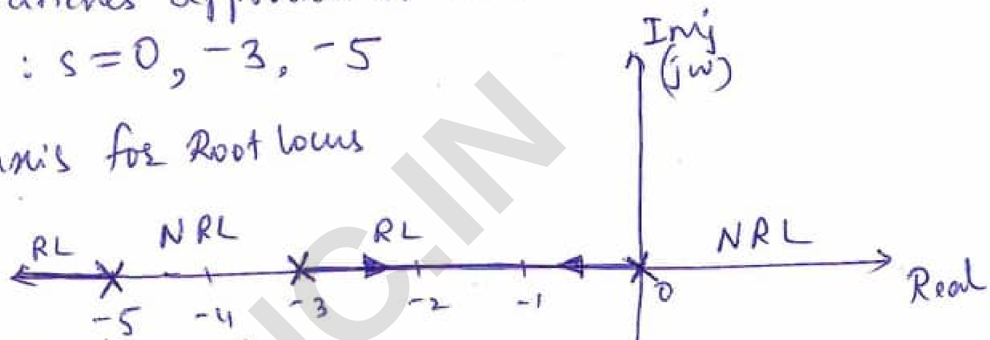
Sol^n: Given: $G(s)H(s) = \dfrac{K}{s(s+3)(s+5)}$

→ Numbers of poles, $P = 3$, $Z = 0$, $N = P = 3$.

Here $P - Z = 3$ branches approach to ∞.

Starting points: $s = 0, -3, -5$

→ Sections of Real axis for Root locus



One breakaway point exists between $s = 0$ and $s = -3$.

→ **Angle of Asymptotes:**

$$\theta = \frac{(2q+1)180^\circ}{(P-Z)}, \quad q = 0, 1, 2.$$

$\therefore \theta_1 = 60^\circ, \quad \theta_2 = 180^\circ$ and $\theta_3 = 300^\circ$

→ **Centroid:** $\dfrac{\Sigma \text{ Real part of Poles} - \Sigma \text{ Real part of Zeros}}{P - Z}$

$$= \frac{0 - 3 - 5 - 0}{3} = \frac{-8}{3} = -2.667$$

→ **Breakaway point:**

$$1 + G(s)H(s) = 1 + \frac{K}{s(s+3)(s+5)} = 0$$

$$\therefore s^3 + 8s^2 + 15s + K = 0$$

i.e $K = -s^3 - 8s^2 - 15s \longrightarrow (1)$

$$\frac{dK}{ds} = -3s^2 - 16s - 15 = 0$$

i.e $3s^2 + 16s + 15 = 0$

$$s = \frac{-16 \pm \sqrt{16^2 - 4(3)(15)}}{2 \times 3} \Rightarrow \underline{s = -1.2137} \text{ and } s = -4.119$$

$\therefore$ $s = -1.2137$ is a valid breakaway point.

Using equation ①, $K = 10.2837$ at $s = -1.2137$

→ Intersection with the imaginary axis:

$$1 + G(s)H(s) = 0$$

$$s^3 + 8s^2 + 15s + K = 0$$

The Routh's array is,

s^3	1	15
s^2	8	K
s^1	$\frac{120-K}{8}$	0
s^0	K	

For Kmax,

$$\frac{120-K}{8} = 0$$

$$K = 120$$

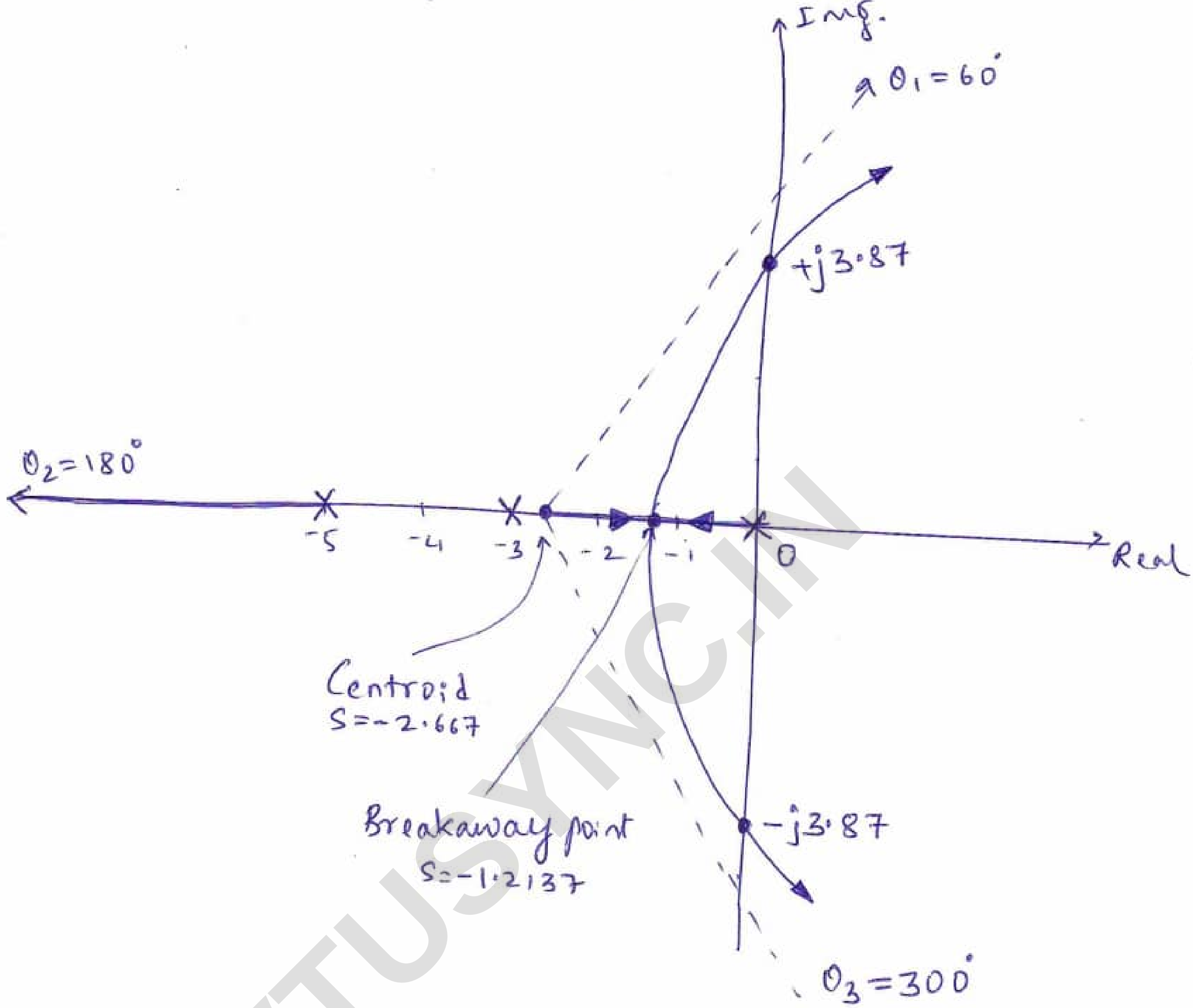
$$A(s) = 8s^2 + K = 0$$

$$s^2 = \frac{-K}{8} = -\frac{120}{8}$$

$$\boxed{\therefore s = \pm j3.873}$$

→ No complex poles or zeros, hence angle of departure is not required.

→ Now drawing the complete Root Locus.



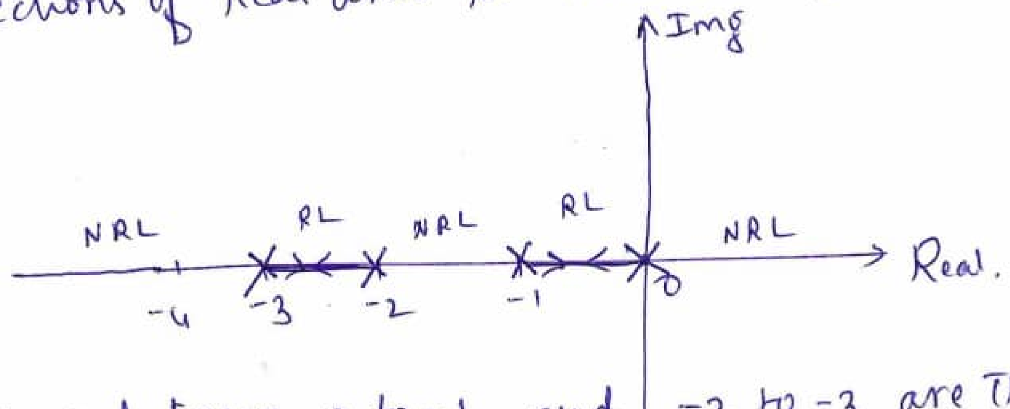
(2) Construct the Root Locus of the system having

$$G(s)H(s) = \frac{K}{s(s+1)(s+2)(s+3)}$$

Soln: Given: $G(s)H(s) = \dfrac{K}{s(s+1)(s+2)(s+3)}$

→ Number of poles, $P = 4$
Number of zeros, $Z = 0$
Number of branches, $N = P = 4$
$P - Z = 4$ branches approach to ∞ with the starting points, $s = 0, -1, -2, -3$.

→ Sections of Real axis for Root Locus,



Sections between 0 to -1 and -2 to -3 are the parts of the Root Locus.
There are two breakaway points.
One between $s = 0$ and -1 and second between $s = -2$ and -3.

→ **Angle of Asymptotes :**

$$\theta = \frac{(2q+1)180^\circ}{P-Z} \; ; \; q = 0, 1, 2, 3$$

$\theta_1 = 45^\circ$, $\theta_2 = 135^\circ$, $\theta_3 = 225^\circ$ and $\theta_4 = 315^\circ$

→ **Centroid :**

$$\sigma = \frac{\Sigma \text{ Real part of poles} - \Sigma \text{ Real part of zeros}}{P-Z}$$

$$= \frac{0-1-2-3}{4} = \underline{\underline{-1.5}}$$

→ **Breakaway point :**

$$1 + G(s)H(s) = 0$$

i.e, $$1 + \frac{K}{s(s+1)(s+2)(s+3)} = 0$$

$$s^4 + 6s^3 + 11s^2 + 6s + K = 0$$

i.e $K = -s^4 - 6s^3 - 11s^2 - 6s \longrightarrow (1)$

$$\frac{dK}{ds} = -4s^3 - 18s^2 - 22s - 6 = 0$$

i.e $4s^3 + 18s^2 + 22s + 6 = 0$

Solving the above equation,

$s = -1.5$, $s = -0.381$ and $s = -2.619$

$\therefore$ $s = -0.381$ and $s = -2.619$ are the valid breakaway points.

Substituting in eq^n (1)

For $s = -0.381$, $K = 1$

& $s = -2.619$, $K = 1$

Both occur simultaneously at $K = 1$.

→ Intersection with the Imaginary axis :

$$1 + G(s)H(s) = 0$$

$$s^4 + 6s^3 + 11s^2 + 6s + K = 0$$

Routh's array is,

s^4	1	11	K
s^3	6	6	0
s^2	10	K	0
s^1	$\frac{60-6K}{10}$	0	
s^0	K		

For Kmax,

$$\frac{60-6K}{10} = 0$$

$\therefore$ $K_{max} = 10$

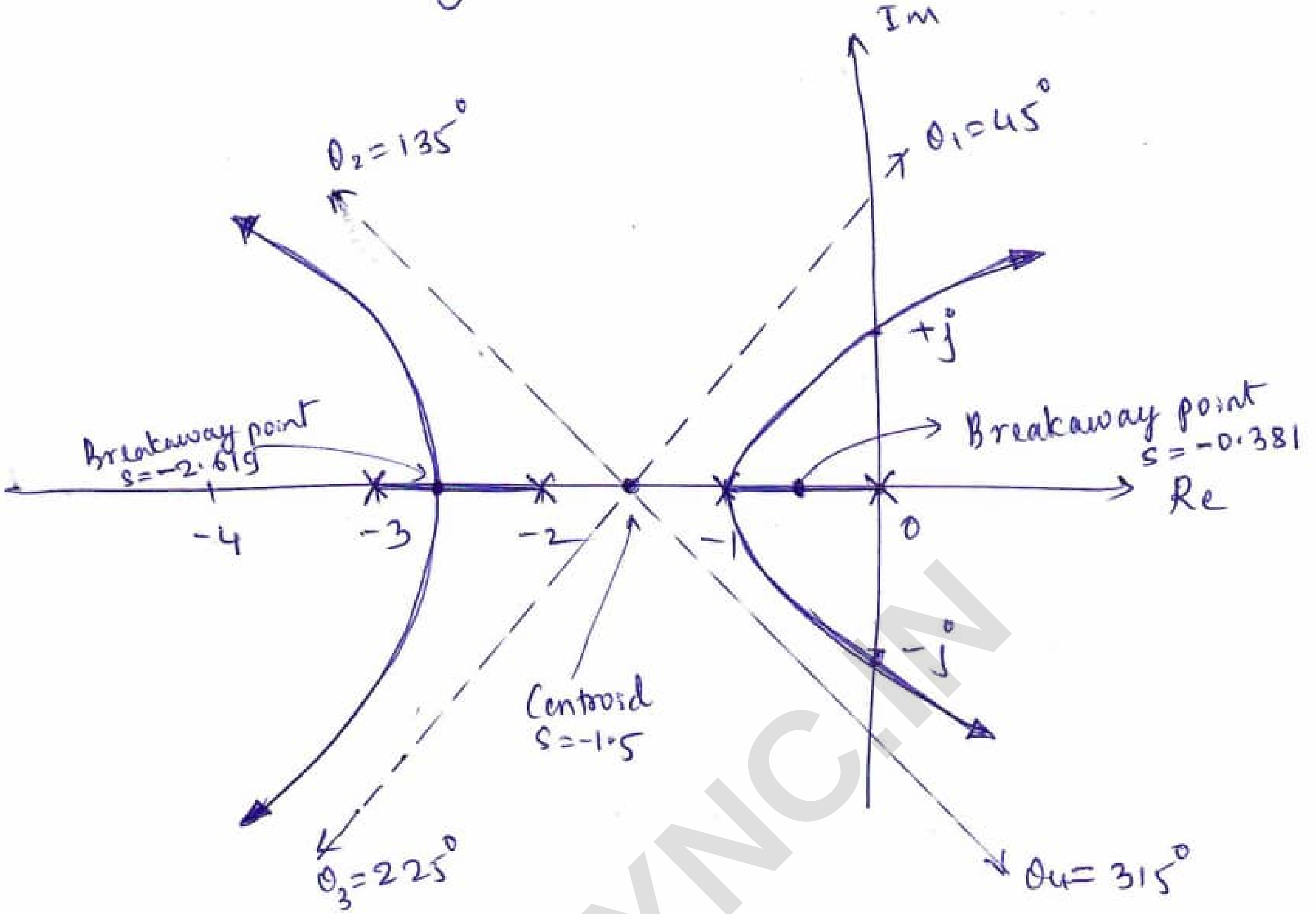
Auxiliary equation:

$$A(s) = \cancel{60-6K=0}$$
$$= 10s^2 + 10 = 0$$
$$s^2 = -1$$

$$\therefore s = \pm j$$

→ No complex poles or zeros, hence angle of departure is not required.

→ Now drawing the complete Root Locus,



(3) A feedback control system has an open loop transfer function $G(s)H(s) = \dfrac{K}{s(s+3)(s^2+2s+2)}$.

Draw the root locus as K varies from $-\infty$ to ∞.

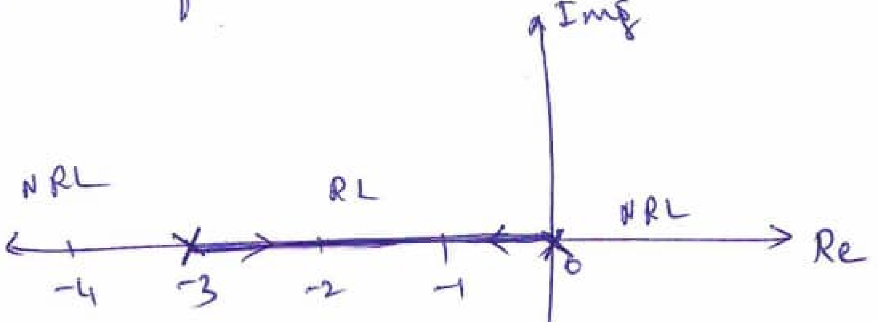
Sol: ⟶ Number of poles, $P = 4$.

Number of zeros, $Z = 0$

Number of branches, $N = P = 4$

$P-Z = 4$ branches approach to ∞ with the starting points, $s = 0$, $s = -3$, $s = -1 \pm j$

→ Sections of Real axis for Root Locus,



There is one Breakaway point between $s=0$ & $s=-3$.

→ <u>Angle of Asymptotes</u>:

$$\theta = \frac{(2q+1)180^\circ}{P-Z} \quad ; \; q = 0,1,2,3$$

$\therefore \; \theta_1 = 45^\circ, \; \theta_2 = 135^\circ, \; \theta_3 = 225^\circ$ and $\theta_4 = 315^\circ$

→ <u>Centroid</u>:

$$\sigma = \frac{\sum \text{Real parts of poles} - \sum \text{Real parts of zeros}}{P-Z}$$

$$= \frac{0-3-1-1-0}{4} = \underline{\underline{-1.25}}$$

→ <u>Breakaway point</u>:

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s(s+3)(s^2+2s+2)} = 0$$

$$\therefore \; s^4 + 5s^3 + 8s^2 + 6s + K = 0$$

i.e $\quad K = -s^4 - 5s^3 - 8s^2 - 6s \quad \rightarrow (1)$

$$\therefore \frac{dK}{ds} = -4s^3 - 15s^2 - 16s - 16 = 0$$

i.e $\quad 4s^3 + 15s^2 + 16s + 16 = 0$

Solving, $s = -2.288$ is a valid breakaway point

from eq^n (1) $\quad \underline{K = 4.3315}$

→ **Intersection with the imaginary axis.**

The characteristic equation is,

$$s^4 + 5s^3 + 8s^2 + 6s + K = 0$$

s^4	1	8	K
s^3	5	6	0
s^2	6.8	K	
s^1	$\frac{40.8 - 5K}{6.8}$	0	
s^0	K		

For Kmax,

$$\frac{40.8 - 5K}{6.8} = 0$$

$$K_{max} = \frac{40.8}{5}$$

$$K_{max} = 8.16$$

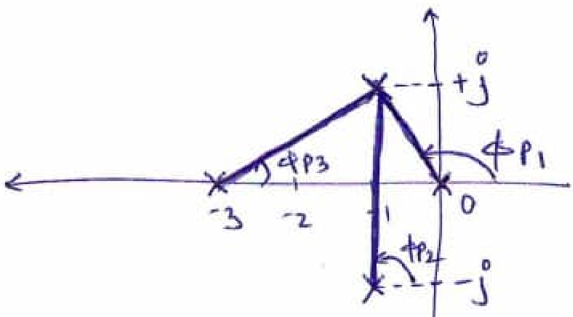
Auxiliary eq^n: $A(s) = 6.8s^2 + K = 0$

$$\Rightarrow 6.8s^2 + 8.16 = 0$$

$$s^2 = -1.2$$

$$\therefore s = \pm j1.095$$

→ **Angle of departure:**



$$\phi_d = 180° - \phi$$

where $\phi = \Sigma\phi_p - \Sigma\phi_z$

$$\phi_{P_1} = 180° - \tan^{-1}\left(\frac{1}{1}\right)$$

$$\phi_{P_1} = 135°$$

$$\phi_{P_2} = 90°$$

$$\phi_{P_3} = \tan^{-1}\frac{1}{2} = 26.56°$$

$$\therefore \Sigma\phi_p = \phi_{P_1} + \phi_{P_2} + \phi_{P_3}$$

$$= 135° + 90° + 26.56°$$

$$\Sigma\phi_p = 251.56°$$

$$\Sigma\phi_z = 0$$

$$\therefore \ \phi = \Sigma \phi_p - \Sigma \phi_z$$

$$\phi = 251.56^\circ$$

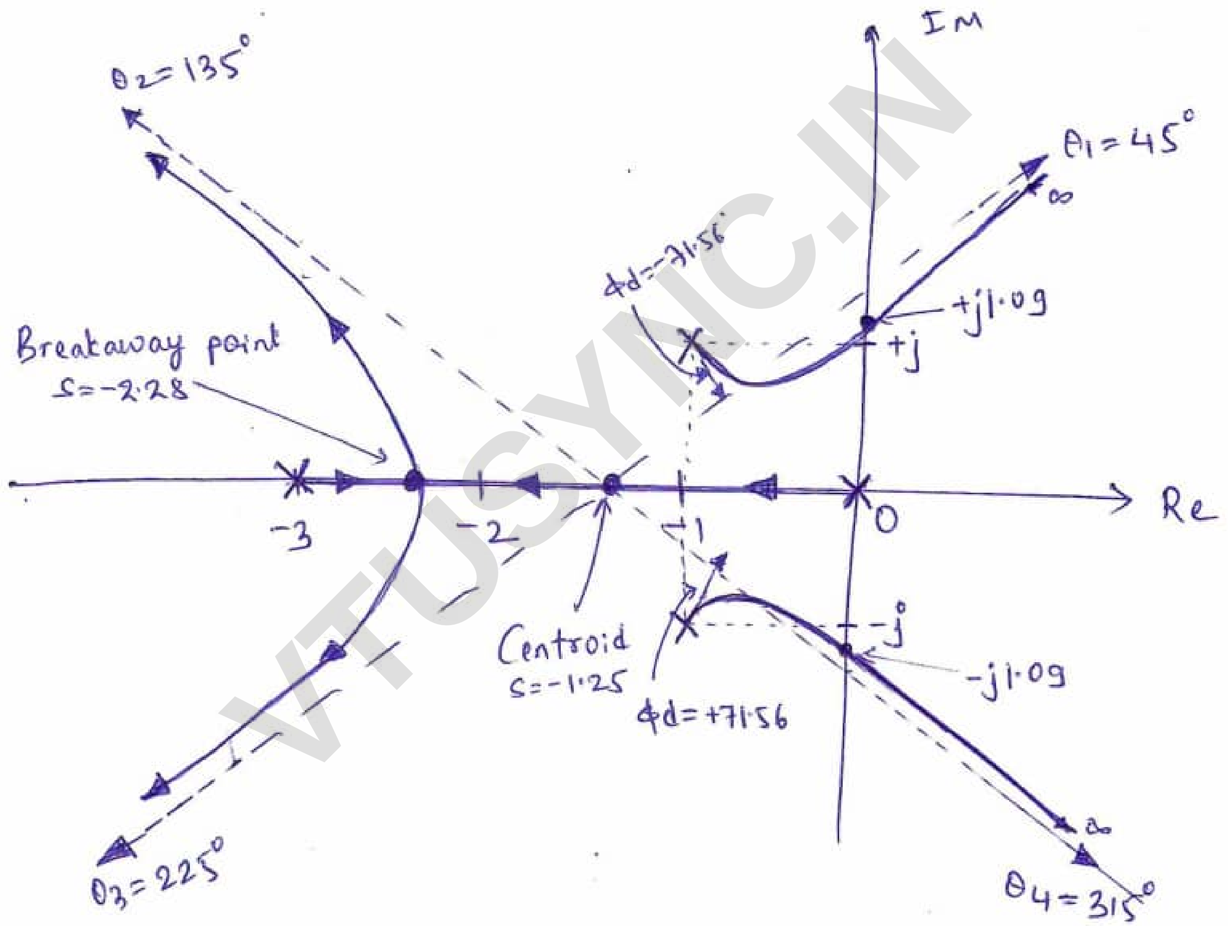
$$\phi d = 180^\circ - \phi$$

$$= 180^\circ - 251.56^\circ$$

$$\phi d = -71.56^\circ \quad \text{at } -1+j$$

$$\text{and } \phi d = +71.56^\circ \quad \text{at } -1-j$$

→ Now drawing the complete Root Locus,



(4) A feedback control system has an open loop transfer function $G(s)H(s) = \frac{K}{s(s+4)(s^2+4s+20)}$

Draw the Root Locus.

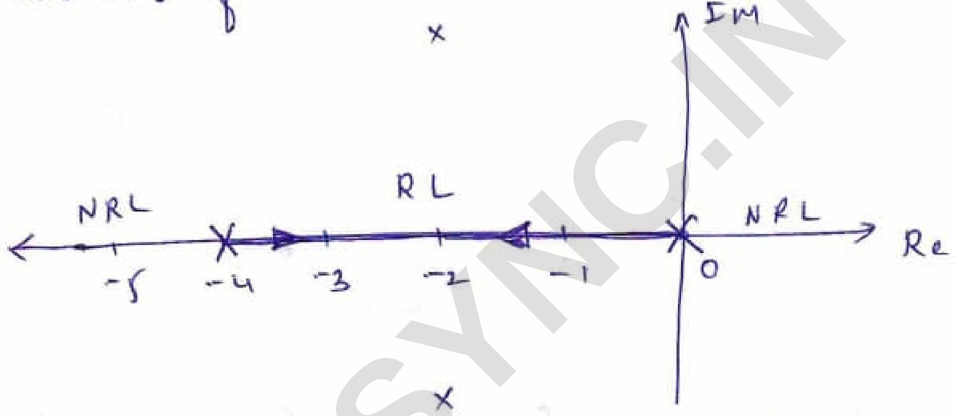
Soln: → Number of poles, $P = 4$

Numbers of zeros, $Z = 0$

Numbers of branches, $N = P = 4$

$P-Z = 4$, branches approach to ∞ with the starting points, $s = 0, -4, -2+j4, -2-j4$

→ Sections of the real axis for Root Locus,



There is one breakaway point between $s = 0$ and $s = -4$.

→ Angle of Asymptotes:

$$\theta = \frac{(2q+1)180^\circ}{P-Z} \quad ; q = 0,1,2,3$$

$\therefore \theta_1 = 45^\circ, \theta_2 = 135^\circ, \theta_3 = 225^\circ$ & $\theta_4 = 315^\circ$

→ Centroid:

$$\sigma = \frac{\Sigma \text{ Real parts of poles} - \Sigma \text{ Real parts of zeros}}{P-Z}$$

$$= \frac{(0-4-2-2)-(0)}{4}$$

$\sigma = -2$

→ Breakaway points:

$$1 + G(s)H(s) = 0$$

i.e $1 + \dfrac{K}{s(s+4)(s^2+4s+20)} = 0$

$\therefore s^4 + 8s^3 + 36s^2 + 80s + K = 0$

i.e $K = -s^4 - 8s^3 - 36s^2 - 80s \rightarrow (1)$

$\therefore \dfrac{dK}{ds} = -4s^3 - 24s^2 - 72s - 80 = 0$

i.e $s^3 + 6s^2 + 18s + 20 = 0$

Solving the eqn,

$s = -2$ and $s = -2 \pm j2.45$

$\underline{s=-2}$ is a valid breakaway point.

from eq (1), $K = 64$.

→ <u>Intersection with imaginary axis</u>

The characteristic eqn is,

$s^4 + 8s^3 + 36s^2 + 80s + K = 0$

The Routh's array is,

s^4	1	36	K
s^3	8	80	0
s^2	26	K	
s^1	$\dfrac{2080-8K}{26}$	0	
s^0	K		

For, Kmar,

$\dfrac{2080-8K}{26} = 0$

$\therefore K_{mar} = 260$

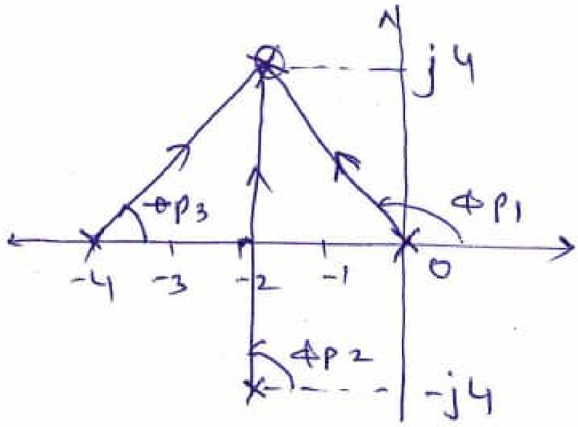
The auxiliary equation is,

$A(s) = 26s^2 + K = 0$

$\therefore s^2 = -10$

$\therefore s = \pm j3.162$

→ Angle of departure :



$\phi p_1 = 180° - \tan^{-1}\left(\frac{4}{2}\right)$

$= 180° - 63.43°$

$\phi p_1 = 116.56°$

$\phi p_2 = 90°$

$\phi p_3 = \tan^{-1}\left(\frac{4}{2}\right)$

$\phi p_3 = 63.43°$

$\therefore \Sigma \phi p = 116.56° + 90° + 63.43°$

$= 270°$

$\therefore \Sigma \phi z = 0$

$\therefore \phi = \Sigma \phi p - \Sigma \phi z$

$= 270°$

$\therefore \phi d = 180° - \phi$

$\phi d = -90°$ at $-2+j4$ and $\phi_d = 90°$ at $-2-j4$

→ Now, drawing the complete Root Locus,